The Impact of School Accessibility on Enrollment: Empirical Evidence from Vietnam

Name: Nguyen Dinh An Hanh
Advisor: Nguyen Thi Hoang Lan

Fulbright University Vietnam

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1 Introduction

In developing countries, governments often prioritize educational policies due to their foundation on human capital theory. Human capital theory states that education acts as a critical investment that enhances individual productivity and drives economic growth. Therefore, education is a powerful tool to reduce poverty, especially in disadvantaged areas. Particularly, Vietnam has made remarkable progress in expanding access to education. In 1993, 15% of children were out of school, followed by a plummet to just 2% in 2014 (Dang and Glewwe (2018)). A similar pattern was observed in upper secondary schools, where the out-of-school rates drastically decreased from 73% to 28% (Dang and Glewwe (2018)).

Despite this achievements, significant challenges still exist, specifically in increasing accessibility to upper secondary education in rural and mountainous areas. While primary school enrollment has seen substantial improvement, enrollment rates in secondary and upper secondary schools remain low in these areas (O'Connell et al. (2022)). This underlines the necessity for persistent effort from government side to assist educational attainment in these under-served communities. Furthermore, current studies often target on a more general public policy overview and aggregate outcomes and there has been little discussion on the effectiveness of each individual intervention on educational outcomes like enrollment rate our cognitive skills. This literature gap, in my opinion, requires more attention and necessitates a deeper examination since this could help governments optimize resource allocation and shape more effective post-expansion education policies.

Vietnam presents a compelling case study in this context. The country is transitioning from a focus on expanding basic access to prioritizing educational quality and retention, particularly in higher education. In this shift, evaluating the impact that increased school accessibility on enrollment rates is critical. Such an analysis can guide future policy decisions: Should investments continue to build more schools to make them more accessible, or should they shift toward improving quality or addressing demand-side barriers like household income and parental awareness? This thesis aims to address this gap by evaluating the causal relationship between school accessibility and enrollment rates in Vietnam. Specifically, it investigates the following research question: *How does improving school accessibility affect student enrollment rates across primary, lower-secondary, and upper-secondary education levels?* This study aims to answer this question by comparing the effects across the three main educational levels (primary, lower secondary and upper secondary) and be able to derive targeted insights for policymakers and researchers. By comparing effects across three levels of education (primary, secondary, and upper-secondary), the study aims to provide targeted insights for policymakers.

The rest of this paper is structured as follows: Focusing on developing countries, Section 2 reviews the existing literature on school accessibility and its impact on education outcomes. Section 3 goes into details about the data sources used in the analysis, including the Vietnam Household Living Standards Survey (VHLSS) and the Vietnam Statistical Yearbook. The empirical approached used to estimate the relationship between school accessibility and student enrollment outcomes is presented in Section 4. Section 5 demonstrates and interprets the results from the regression models, and the thesis is concluded with a summary of the main findings and implications for education policy in Section 6.

2 Literature Review

2.1 Vietnamese Educational Structures

Vietnam's formal educational system for consists of four levels: preschool, primary, lower secondary, and upper secondary. Preschool is an optional stage opened for children aged from 3 to 5, offered by both public and private institutions. At age 6, children are qualified for primary education. Primary education is a compulsory stage that requires 5 years to complete (Grades 1–5). It offers students foundational subjects like Vietnamese language, mathematics, social studies, physical education, music, and art. The lower secondary level (Grades 6–9) is established on these foundations, with the addition of subjects such as natural sciences, foreign languages, and computer science. Upper secondary level (Grade 10–12), on the other hand, is optional and may provide specialized tracks in natural sciences, technology, social sciences, or foreign languages. The official age for each level of education are: 6-10 years old (the primary age group); 11-14 years old (the lower secondary age group); population aged 15–17 years (the upper secondary age group). All four levels of the formal educational system mentioned above are overseen by the Ministry of Education and Training (MOET), while vocational education and training (VET) falls under the administration of the Ministry of Labour, Invalids, and Social Affairs (MOLISA).

2.2 Student Enrollment

Vietnam faces significant challenges in guaranteeing universal education access, particularly for disadvantaged groups in remote and ethnic minority areas. To combat this issue, the government has implemented various education interventions since the early 2000s. Using policies listed in the UNICEF (2013), I categorized them into supply-side and demand-side policies and matched them with existing studies in Vietnam (Table 2.1). According to Glewwe et al. (2020), demand-side policies aim to increase households' willingness or ability to send children to school by lowering financial barriers (e.g., cash transfers, fee reductions) or altering negative perceptions of education's value (e.g., by information campaigns). On the contrary, supply-side policies focus on improving the availability or quality of schooling services, such as constructing schools, hiring teachers, or providing learning materials—variables.

Table 2.1: Education Interventions in Vietnam, Categorized by Glewwe Framework

Policy Category	Studies	Vietnam-specific Policy	Decrees or Programs	Impact on Enrollment
Demand-side Policies				
Unconditional Transfers	Bui et al. (2020) Khiem et al. (2020)	Provides cash support for very small ethnic minority groups living in poor households. The cash support varies for different groups.	Decision No. 2123/QD-TTg (22 Nov 2010)	Yes
	N/A	Regulates rice subsidies of 15 kg per month for nine months for students in extremely disadvantaged areas	Decision 36/2013/TTg (18 June 2013)	N/A

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Table 2.1 – *Continued from previous page*

Policy Category	Studies	Vietnam-specific Policy	Decrees or Programs	Impact on Enrollment
Conditional Cash Transfers	Bui et al. (2020) Khiem et al. (2020)	Provides support for upper secondary students in difficult conditions, offering 50% of the average minimum salary for nine months/pupil	Decision No. 12/2013/QD-TTg (24 Jan 2013)	Yes
Merit-Based Scholarships	N/A	Adjusts government scholarships for ethnic minority students at boarding schools and pre-university schools.	Decision No. 82/2006/QD-TTg (14 Apr 2006)	N/A
		Provides priority scholarships to students.	Decision No. 152/2007/QĐ-TTg (14 Sep 2007)	
		Supports educational expenses for ethnic minority students in universities.	Decision No. 66/2013/QD-TTg	
Supply-Side Policies				
School Construction	N/A	Addresses deteriorating and temporary school maintenance in remote and ethnic minority areas	Directive No. 02/CT-TTg (22 Jan 2013)	N/A
Teacher Attendance	N/A	Provides incentives for teachers and management staff in extremely disadvantaged areas	Decree 61/2006/ND-CP (20 June 2006)	Yes
Teacher Incentives	Nguyen (2006)	Implements incentive policies for teachers and education managers in remote, ethnic minority areas	Directive No. 02/CT-TTg (22 Jan 2013)	N/A
Material Inputs (Textbooks, Supplies)	Nguyen (2006)	Supports disadvantaged children, including ethnic minorities, by providing textbooks and learning materials	Decision No. 62/2005/QD-TTg (24 Mar 2005)	Yes

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Table 2.1 – *Continued from previous page*

Policy Category	Studies	Vietnam-specific Policy	Decrees or Programs	Impact on Enrollment
	Nguyen (2006)	Supports nominations for teacher training programs with learning materials provided to students Decree No. 134/2006/ND-CP (14 Nov 2006)	N/A	
	N/A	Provides learning materials for students with disabilities, including sign language resources	Decision No. 1019/QD-TTg (5 Aug 2012)	N/A

Bui et al. (2020) and Khiem et al. (2020) studied the effect of two demand-side interventions: tuition fee reduction and education subsidy program, but with different approaches. In Bui et al. (2020) research, they used logistics linear regression to identify correlation and evaluated the impact of the two policies on primary school students. The results showed that tuition fee reduction is associated with an increase by 5.3% in the enrollment rate while education subsidy raised it by 6%. These incentives were also proven to be considerably effective in ethnic minority groups using interaction terms, with the tuition fee reduction contributing 25 percentage points and the education subsidy growing 12 percentage points for the enrollment rate in this particular group.

However, Khiem et al. (2020) adopted the difference-in-differences method to establish a causal effect on three educational levels in Vietnam: primary, lower secondary, and upper secondary. Their main focus is on the 2010 policy, which exclusively supports poor children. Since the pre-existing differences between the two groups potentially violate the parallel trend assumption, the authors strengthen the approach by using propensity score matching. Importantly, the effect considered in this paper is from education subsidies only. Because for the primary level, poor children's tuition fees are waived due to the effect of a 1998 policy, while for the lower secondary level, both poor and non-poor children received the same 50% reduction in tuition fees. The findings indicated that, for primary-level enrollment, receiving education subsidies increased the enrollment rate of poor children by 1.2 percentage points, while for lower secondary-level enrollment, their enrollment rate rose by 2.2 percentage points. However, there are no statistically significant differences in the enrollment rate at the upper secondary level.

Meanwhile, Nguyen (2006) finding indirectly suggested that there is no significant result of tuition fees on enrollment. She analyzed Vietnam Household Living Standards Survey data from 1997–1998, to examine how school and household characteristics affect student enrollment in rural areas. Using logistic regression, Nguyen's paper identified a correlation between school quality indicators (student-teacher ratio, classrooms with good blackboards, teacher qualifications, etc) on enrollment rate. Among the mentioned variables, the addition of a classroom with a good blackboard and teacher qualifications each contributed 1% to enrollment, and other variables showed no significant effect on enrollment.

A more comprehensive intervention, the Escuela Nueva (VNEN) program, was evaluated by Dang et al. (2022). VNEN is a multi-faceted education reform initiative in Vietnam that aims to improve teaching methods, classroom resource, student collaboration, and parental engagement. Using propensity score matching and OLS methods, Dang et al. (2022) found modest short-term gains in both cognitive and non-cognitive skills for primary students, with stronger effects for ethnic minority groups. However, these gains diminishes over time, raising concerns about sustainability. The mixed results on gender effects further complicate the narrative. Unlike more demand-side focusing interventions, comprehensive reforms like VNEN may require continuous development iterations to sustain long-term influence and improvement in the long run.

2.3 School Construction

2.3.1 Overview and Organization

The school construction program was chosen due to its strong correlation with accessibility enhancement for students. School construction here refers to government- or donor-funded initiatives meant to build new schools or upgrading existing educational infrastructure to lower physical barriers to schooling. These efforts are specifically focused on underserved, rural, or remote communities, where long travel distances, terrain and poor facilities are often the limiters to educational accessibility.

In this subsection, I categorize the outcomes of school construction into two key dimensions: short-term effects—such as changes in enrollment and academic performance—and long-term effects, including educational attainment, labor market participation, and broader socio-economic outcomes.

2.3.2 School Construction's Outcome

Short-term Outcome Several studies, such as Handa (2002), Alderman et al. (2003), Burde and Linden (2013), Kazianga et al. (2013), etc., have been carried out to examine the effect of school construction on several aspects, such as student enrollment and test scores. Handa (2002) was one of the first to examine the student enrollment outcomes for children in the primary school (aged 7–11). The linear regression analysis reveals that an additional school constructed in "administration post" increased boys' enrollment by 0.2%, though it had no effect on girls' enrollment. The results of the difference-in-differences regression show a similar pattern: the school enrollment rate is positive and significant for the full sample (0.3 percentage points); for boys, it is even much higher (0.4 percentage points), but it remains insignificant for girls. Handa (2002)'s policy simulation results are also noteworthy as he predicted the policy's effectiveness, suggesting that constructing additional schools or improving the adult literacy rate will exert a more significant influence on primary school enrollment rates compared to interventions aimed at increasing household income. This is one of the key highlights and contributions because later literature rarely compares different interventions to see their effectiveness and provide recommendations for policymakers.

Building up on this discussion, Alderman et al. (2003) investigate the effectiveness of two fellowship programs targeting urban and rural girls in Pakistan. The implementation of this program aimed to stimulate the establishment of private schools for impoverished girls. Using randomized control, findings show that after considering other factors, the urban fellowship program in 1995 resulted in the treatment group outperforming the control group by 22.4% for boys and 33.4% for girls. On the other hand, the rural fellowship program experienced a decrease of -10.1% for boys and an increase of 15.2% for girls.

In alignment with the paper, Burde and Linden (2013) examined a policy implemented in Afghanistan since 2007. The findings also indicate that the presence of a school in the village increased enrollment by 52 percentage points for girls and 35 percentage points for boys, respectively. Moving forward to the U.S., Neilson and Zimmerman (2014) also established a causal relationship between public school construction and student enrollment for three levels in New Haven, Connecticut, from 1998 to mid-2010. Using difference-in-difference methods, his findings indicate that after six years post-construction, student enrollment effects reach 17.3 percentage points. In contrast, Lavy and Nixon (2017) found that there is no effect of school construction on enrollment and attendance rates. This variation could be attributed to the paper's scope being slightly different, as it assessed rebuilt elementary schools in Texas, U.S., unlike other cases that focused on newly built schools. The only significant result in Lavy and Nixon (2017)'s paper is student outcomes. However, Lavy and Nixon (2017)'s paper assessed it using the state percentile ranking, not test scores, which is a prevalent metric in most current studies.

Using multiple regression analysis to indicate correlation only, Burde and Linden (2013) paper studied these outcomes in the same sense. She found out that average test scores for girls and boys in spring 2008, which is one year after the school construction intervention, increased by 0.66 and 0.41 standard deviations, respectively. The same patterns were observed in the Kazianga et al. (2013) paper, where the test score for exams covering math and French improved by 0.41 standard deviations for all children and increased by 2.2 standard deviations for children attending school due to the school construction program. Unlike these results, in the Neilson and Zimmerman (2014) paper, reading scores increase by 12.1 percentage points, significant at the 10% level only, and the math score pattern is insignificant. Taken together, these papers provide a comprehensive understanding of the short-term and educational effect on student enrollment with this supply-side school construction.

Long-term Outcome While some researchers concentrate on short-term outcomes, such as student enrollment and student achievements, others extend their focus to long-term outcomes, such as education attainment (Duflo (2001)), tax revenues, and living standards (Akresh et al. (2023)), fluctuations in home prices (Christopher et al. 2014), and the provision of public goods (Martinez-Bravo (2017)). Duflo (2001) employs quasi-experimental difference-in-differences (DID) methodology, merging data of males who were born in 1950-1972 from an intercensal survey with data from Indonesia's major school construction program in the 1973-1978 periods. It concludes that the construction of an average of 1 school per 1,000 children will enhance their educational attainment from 0.12 to 0.19—this result is specifically for males.

Following up on the educational findings and grounding of Duflo (2001)'s work, Akresh et al. (2023) extend their focus by examining the impact of school construction on females and observing gender differences for both first- and second-generation outcomes. Among the first generation—individuals who were directly exposed to the school construction program—years of schooling increased by 0.23 for females and 0.27 for males. For the second generation, if their mother had experienced this program, their years of schooling would increase by 0.17 years, while father's engagement to the program had no significant effects. Besides education outcomes, Akresh et al. (2023) also expanded the scope by examining labor market outcomes, marriage market outcomes, and household expenditure outcomes. Regarding the labor market, which included current employment, agricultural jobs, and migration likelihood, male exposure had a significant impact by enhancing formal employment by 0.6 percentage points, reducing the likelihood of employment in agricultural jobs by 1.2 percentage points, and increasing the likelihood of migration by 0.7 percentage points. Females, on the other hand, only experience a rise in migration likelihood (0.8 percentage points). Turning to the marriage and household expenditure outcomes, female exposure increased the education level of spouses by 1.12 years, while male exposure resulted in a larger gain of 1.18 years. Total household expenditures increased significantly for female-exposed households (2.5 percentage points). In contrast, male exposure had no significant impact on these outcomes.

Expanding the scope of inquiry, Martinez-Bravo (2017) analyzes the effect of school construction on public provision. He also examined the Indonesian program using the same setup as Duflo (2001) and Martinez-Bravo (2017). For marginal effects, each additional school constructed in a village contributes to an increase in the number of primary health centers (1.2 percentage points), access to safe water (1.8 percentage points), and the presence of doctors (0.6 percentage points). When looking at villages that have one more INPRES school than the average, these communities saw increases of 7 percent in primary health centers, 6 percent in access to safe water, and 1.7 percent in the number of doctors compared to their average levels. Neilson and Zimmerman (2014) found that when controlling for other variables, housing prices increased by 5.4% after the interventions. This result is significant at the 5% level. However, if we separate the analysis into specific stages (filling/construction), the results for each stage are not significant. In conclusion, school construction programs yield significant long-term benefits. They improve educational attainment, labor and marriage market outcomes, house prices, and even public provision.

2.3.3 This paper's contribution

Firstly, this thesis makes several contributions to the understudied field of education policy in Vietnam, where empirical research remains scarce due to data limitations, fragmented documentation, and a lack of standardized longitudinal datasets. Despite these challenges, this study deliberately focuses on Vietnam to provide basic evidence for future research, acting as an initial analysis that tests different models and creates variables to look for strong patterns.

Furthermore, this study helps fill an important gap in research on education in Vietnam. Most of the existing studies only focus on financial policies—such as scholarships or cash transfers—that makes schooling more affordable (demand-side). However, there has been few studies examining how building more schools (supply-side) affects enrollment, especially differences across provinces and over time.

Finally, as one of the first systematic attempts to analyze school accessibility effects across multiple education levels in Vietnam, this thesis intentionally adopts an experimental approach—testing variable constructions and model specifications—to document data limitations and propose possible enhancements for future research. This approach aims to establish a foundation for a more nuanced and data-driven education policy discussion in Vietnam.

3 Data

3.1 Net Enrollment Rate

The net enrollment rate (*"tỉ lệ đi học đúng tuổi"*) serves as the main outcome variable in this study. It is defined as the number of children within the official school age who are enrolled in the corresponding level of education, divided by the total number of children in that age group. In comparison, the gross enrollment rate (*"tỉ lệ đi học chung"*) counts all students enrolled at a given education level, regardless of their age, and divides that by the number of children in the official age group for that level Dang and Glewwe (2018). The net enrollment rate definition given by the General Statistics Office (GSO (2023)) also matches this explanation. In Vietnam, the associated ages for each level of education are age 6-10 (primary), age 11-14 (lower secondary), and age 15-17 (upper secondary).

The VHLSS is used to calculate the net enrollment rate at the provincial level, assuming that the sample estimates are close to the unknown population parameter values. The VHLSS provides annual survey data on approximately 40,000 households across 63 provinces, recording information at the individual level. It includes key demographic characteristics related to living standards and socio-economic indicators such as education, health care, employment, income, consumption expenditure, durable goods, et cetera. The survey is conducted biennially and was administered over four rounds each period, correspondingly to 4 different quarters of the year. Data were collected through face-to-face interviews with household heads, household members, and key commune officials. This study uses 8 waves of the VHLSS: 2006, 2008, 2010, 2012, 2014, 2016, 2018, and 2020. Given the information about household members' ages and their enrollment status (yes/no), the net enrollment rate was constructed covering three levels of general education: primary, lower secondary, and upper secondary.

3.2 School accessibility

To evaluate the impact on school accessibility, this paper selected three potential explanatory variables: (1) number of schools, (2) number of schools per 1,000 people, and (3) number of schools per 1,000 hectares of homestead land. These variables are derived from the Vietnam Statistical Yearbook, and a detailed explanation of the calculation methods is presented in Table 3.2. Similar to the VHLSS, the Statistical Yearbook also offers a comprehensive annual dataset on the general socio-economic dynamic and situation of Vietnam, but it is worth noting that these data are recorded at a population level.

When merging data from the Vietnam Statistical Yearbook with the VHLSS, a difference in reference periods becomes evident. The homestead land area published in the Statistical Yearbook reflects the data of the previous year, while the VHLSS reflects data collected throughout that year, from the first to the fourth quarter. To address this temporal misalignment, this study used homestead land data one year ahead to correspond more closely with the number of school and enrollment data.

Table 3.1: Variables Description

Variables	Description	Construction Formula	Type
Dependent variables			
$enrollment_{i,t}$	Net enrollment rate in province i at year t	$enrollment_{i,t} = \frac{enrolled_{i,t}}{school\_age_{i,t}} \times 100\%$	Sample
Independent variables			
$schools_{i,t}$	The total number of schools in province i at year t	Data comes from Statistical Yearbook	Population
$schools\_pc_{i,t}$	The number of schools per 1000 residents in province i at year t	$schools\_pc_{i,t} = \frac{schools_{i,t}}{\bar{P}_{i,t}}$	Population
$schools\_psa_{i,t}$	The number of schools per 1000 ha in province i at year t	$schools\_psa_{i,t} = \frac{schools_{i,t}}{homestead_{i,t}}$	Population
μ_t	year fixed effect		
ω_i	provincial fixed effect		
$\varepsilon_{i,t}$	idiosyncratic error term		
Base variable for calculation			
$enrolled_{i,t}$	Number of students enrolled at certain level of education for province i in year t	Data counts from the VHLSS	Sample
$school\_age_{i,t}$	Number of students in the official age group at certain level of education for province i in year t	Data counts from the VHLSS	Sample
$\bar{P}_{i,t}$	The average population of province i in year t	Data comes from Statistical Yearbook. Since the GSO may collect population data at multiple points in time, the reported figures are the average of these values.	Population
$homestead_{i,t}$	The homestead land area in province i, at time t.	Data comes from Statistical Yearbook.	Population

4 Methodology

In this paper, I run three sets of regression models to evaluate the effects of school accessibility on enrollment rates. The first model uses the number of schools as independent variables; the second uses schools per capita; and the last one uses schools per area as independent variables. I employ these three measures to explore which best approximates school accessibility. While more accurate indicators—such as travel distance or commuting time—exist in the literature, such data are currently not available in the Vietnamese setting. Therefore, I rely on these variables as practical alternatives to perform my analysis.

I also apply two complementary approaches. The first is a two-way fixed effects (2FE) regression model, which controls for both province-level fixed effects and year-specific fixed effects. It is intuitive to assume that the variations in our observations are not random. Therefore, by applying two-way fixed effects, the model can handle unobserved, time-invariant provincial characteristics (e.g., geography, long-standing local policies) and nationwide time shocks (e.g., education reforms or economic fluctuations). This reduces omitted variable bias and focuses on within-province variation, allowing us to assess how changes within each province over time impact the dependent variable. The primary regression equations are specified as:

1. $\text{enrollment}_{i,t} = \beta_0 + \beta_1 \times \text{schools}_{i,t} + \mu_t + \omega_i + \varepsilon_{i,t}$
2. $\text{enrollment}_{i,t} = \beta_0 + \beta_1 \times \text{schools\_pc}_{i,t} + \mu_t + \omega_i + \varepsilon_{i,t}$
3. $\text{enrollment}_{i,t} = \beta_0 + \beta_1 \times \text{schools\_psa}_{i,t} + \mu_t + \omega_i + \varepsilon_{i,t}$

The proposed models and analytical approach will be applied sequentially to each of the three educational levels: primary, lower secondary, and upper secondary.

5 Results

5.1 Descriptive Statistics

Table 5.1 demonstrates the descriptive statistics for both independent and dependent variables. The average enrollment rate decreases, and as education levels go up, variation increases . In Figure A.1, among the three, the primary level has the overall highest enrollment rate, which increases significantly over time. This trend is likely connected to primary education being the foundation for children's learning, and it is also associated with literacy level, making the government to invest more in this age group compared to other educational levels. On the other hand, upper secondary education shows the greatest variation in enrollment rates and lower enrollment rates. As shown in Figure A.2, while most regions show relatively stable and high enrollment levels, there are noticeable disparities. The Red River Delta consistently leads with the highest enrollment rates, getting closer to full participation over the period. In contrast, the Mekong River Delta and Central Highlands lag behind, exhibiting greater volatility and lower enrollment rates. However, both regions demonstrate substantial progresses over time, narrowing the regional enrollment gap by the end of the period.

In Figure A.3, the distribution of the number of schools remains relatively consistent over time and is strongly right-skewed for all three levels of education. The persistence of extreme outliers—likely large, urban provinces such as Hanoi or Ho Chi Minh City—is clearly seen throughout the period. Looking at Figure A.4, the number of schools does not have significant variations and remains quite stable, with a decreasing trend for both primary and lower secondary schools and a slight increase in upper secondary schools. Northern Midlands experienced the most significant drop in the number of schools among 6 regions. This decreased enrollment could be attributed to the merging policy of the government by merging the single-unit school into inter-level schools for Hoa Binh TTXVN (2018) and Lao Cai in 2018 VOV (2018).

Table 5.1: Descriptive Statistics

Variables	Mean	Std. Dev.	Min	Max
Dependent variables				
School enrollment (Primary ages 6–10)	97.38	3.39	82.86	100.00
School enrollment (Secondary ages 11–14)	93.51	6.53	66.67	100.00
School enrollment (Highschool ages 15–17)	73.12	14.82	30.77	100.00
Independent variables				
Primary				
Number of schools	232.69	119.05	29.00	756.00
School per capita (per thousand people)	0.19	0.07	0.03	0.47

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Table 5.1 – *Continued from previous page*

Variables	Mean	Std. Dev.	Min	Max
School per area (per thousand ha)	25.48	10.47	2.03	68.82
Lower Secondary				
Number of schools	156.15	102.92	27.00	652.00
School per capita (per thousand people)	0.13	0.06	0.03	0.32
School per area (per thousand ha)	16.46	7.33	1.89	56.36
Upper Secondary				
Number of schools	36.45	25.35	6.00	195.00
School per capita (per thousand people)	0.03	0.01	0.01	0.06
School per area (per thousand ha)	3.70	1.21	0.99	7.67
No. of Observations		505		

5.2 Regression Results

5.2.1 Effects on primary enrollment rate

Table 5.2 reports the influence of school accessibility on primary enrollment rate. The first column presents results from a pooled OLS model. Province fixed effects in the second column is used to control time-invariant differences across provinces. The third column incorporates time fixed effects to account for changes that affect all provinces equally over the time period. Lastly, the final column includes both province and time fixed effects to capture unobserved heterogeneity across space and time. Each row in the table represents a separate regression: row (1) chooses the number of schools as the independent variable, row (2) uses the number of schools per capita, and row (3) uses the number of schools per area. As such, the estimated effects of each variable should be interpreted individually.

For primary education, none of the three measures of school availability—number of schools, schools per capita, or schools per area—show robust or consistent associations with enrollment when controlling for both province and time fixed effects. In the first row, the coefficient on the number of schools is generally not statistically significant, except in the entity fixed effects model, where it is significant at the 10% level. Therefore, we cannot conclude that the number of schools alone significantly impacts the primary enrollment rate. On the other hand, both schools per capita and schools per area show statistically significant negative associations with enrollment in the base, province FE, and time FE models (at the 1% level), but these relationships disappear once two-way fixed effects are included. With two-way fixed effects, the only variation left is idiosyncratic (specific to an entity and a time period). Since the results are insignificant, this might suggest that the remaining variation might be too weak or noisy to detect an effect.

Interestingly, the magnitude of the negative coefficients for schools per capita and schools per area in the initial models (approximately 14–25% and 0.09–0.17%, respectively) suggests a positive case of diminishing returns. Additional schools may have been constructed in areas where enrollment was already high or where other constraints—such as poverty or cultural norms—continued to limit school participation. From the descriptive statistics, the primary enrollment rate is already close to 100%. As a result, building more schools may have a limited effect in drawing in the remaining out-of-school children. Furthermore, inefficiencies in how schools are distributed could lead to higher operational costs and diluted quality, which may decrease the enrollment rate.

Table 5.2: Regression model results for child enrollment at primary school level (ages 6–10).

	Base Model	Entity FE Model	Time FE Model	Two-way FE Model
(1) schools$_{i,t}$	0.0008	-0.0112*	0.0015	-0.0008
	(0.0015)	(0.0058)	(0.0013)	(0.0029)
(2) schools_pc$_{i,t}$	-14.637***	-25.205***	-11.664***	-3.7337
	(2.8948)	(4.8536)	(2.9419)	(4.0108)
(3) schools_psqkm$_{i,t}$	-0.1119***	-0.1775***	-0.0901***	-0.0575
	(0.0263)	(0.0278)	(0.0280)	(0.0330)
Fixed Effects	None	Entity	Time	Entity + Time
Observations	504	504	504	504

Notes: Each row represents a separate regression. Standard errors in parentheses; *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

5.2.2 Effects on lower secondary enrollment rate

At the lower secondary level, the number of schools is positively and significantly associated with enrollment in the base and time fixed effects models (coefficients of 0.0166 and 0.0169, respectively, significant at the 1% level). However, this association vanishes in the entity and two-way fixed effects models, becoming statistically insignificant or even negative. This mirrors the pattern observed at the primary level, where any apparent effects are largely driven by between-province differences rather than within-province changes over time. The schools-per-capita variable is significant and negative only in the entity fixed effects model and insignificant in other specifications, making it less robust than in the primary case. The schools-per-area variable is not statistically significant in any of the four models for lower secondary enrollment.

Table 5.3: Fixed-effects regression model results for child enrollment at lower secondary school level (ages 11–14).

	Base Model	Entity FE Model	Time FE Model	Two-way FE Model
(1) $\text{schools}_{i,t}$	0.0166***	-0.0077	0.0169***	-0.0046
	(0.0044)	(0.0073)	(0.0045)	(0.0054)
(2) $\text{schools\_pc}_{i,t}$	5.6392	-26.832***	9.0687	-4.4558
	(10.160)	(12.110)	(10.606)	(15.753)
(3) $\text{schools\_psqkm}_{i,t}$	-0.0019	-0.1010	0.0322	0.0955
	(0.0682)	(0.0953)	(0.0724)	(0.1038)
Fixed Effects	None	Entity	Time	Entity + Time
Observations	504	504	504	504

Notes: Each row represents a separate regression. Standard errors in parentheses; *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

5.2.3 Effects on Upper Secondary Enrollment Rate

For upper secondary education, a similar pattern emerges. The number of schools is positively and significantly associated with enrollment in the base and time fixed effects models (coefficient around 0.20, significant at the 1% level), but the effect becomes insignificant once province or two-way fixed effects are included. The estimates for schools per capita and schools per area are inconsistent and not robust across specifications. While some intermediate models (e.g., the entity fixed effects model) show significant results, none of the variables remain statistically significant once both province and time fixed effects are applied. These findings, along with possible measurement and methodological limitations, are further discussed in Section 6.

Table 5.4: Regression model results for child enrollment at the upper secondary level (ages 15–17).

	Base Model	Entity FE Model	Time FE Model	Two-way FE Model
(1) $\text{schools}_{i,t}$	0.2018***	-0.0207	0.1986***	-0.1556*
	(0.0468)	(0.0617)	(0.0464)	(0.0851)
(2) $\text{schools\_pc}_{i,t}$	122.55	-328.18***	144.53	-206.28
	(185.87)	(149.52)	(191.61)	(143.42)
(3) $\text{schools\_psqkm}_{i,t}$	0.3028	-1.7434*	0.5019	-0.7283
	(1.1548)	(0.9050)	(1.1905)	(0.7264)
Fixed Effects	None	Entity	Time	Entity + Time
Observations	504	505	504	504

Notes: Each row represents a separate regression. Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

6 Discussion

Although none of these independent variables fully captures the essence of my thesis, which aims to examine the correlation between enrollment rates and school accessibility, the paper demonstrates a substantial effort in exploring different model specifications. Limitations persist due to several methodological issues. Future research should address the following concerns to improve the robustness and accuracy of the findings:

6.1 Measurement Error

6.1.1 Net Enrollment Rate

The net enrollment rate used in this study is calculated from the Vietnam Household Living Standards Survey (VHLSS), which is based on sample data, while the independent variable is drawn from population-level data. This procedure raises concerns about the sampling adequacy, as I cannot verify whether the VHLSS sample properly reflects provincial-level population traits. Future research should either (1) develop methods to validate sample representativity using available data or (2) obtain detailed government demographic records to apply suitable population-weighting adjustments.

6.1.2 Number of Schools

The use of school quantity as a metric for school accessibility is a major constraint in this study. While intuitive and publicly available, this variable is susceptible to several measurement errors that may create systematic bias, thereby affecting the robustness of the estimated effects.

To begin with, urban-rural heterogeneity complicates the interpretation of the number of schools. In urban areas, private schools often serve a substantial share of students, particularly at secondary and upper secondary levels. Because the dataset primarily includes public schools, its use may lead to systematic undercounting of actual school supply in urban regions.

In addition, the measure does not account for variation in school size or capacity. For example, in two provinces may have the same number of schools, but in case one of them has larger schools with more classes, leading to higher student intake, its actual schooling capacity is significantly higher. Ignoring this variation again causes systematic measurement error, potentially attenuating the true effect of school access. This omission could also obscure the diminishing returns argument: building smaller schools in already well-served areas may show little to no impact on enrollment not because ”schools don’t help,” but because the real capacity increase is minimal.

Finally, the method of counting schools from the data doesn’t reflect cases where different education levels are merged into a single school. For example, in 2016, the provinces of Hoa Binh and Lao Cai implemented a policy that combined primary and lower secondary schools into one shared school. As a result, what used to be recorded as two separate schools now appears as just one in the dataset. This change happened only in these two provinces and in a specific year, so it falls outside what the model can adjust for using time or province fixed effects. It results in a sudden decline in the recorded number of schools—especially noticeable in the Northern Midlands region (see Figure A.4 in the Appendix). This unexpected change adds a lot of variation within provinces that is not tied to real changes in school availability. The model then tries to explain this drop as if it were a meaningful shift, even though it is actually a side effect of a policy that changes how schools are counted. This approach makes it harder to interpret the results accurately.

6.1.3 Schools Per Population

Due to data limitations, this study relies on average total population for each province rather than age-specific population figures. Such an approach may introduce bias in the estimated effects of schools per capita on enrollment rates. The direction of this bias depends on demographic composition across provinces. If provinces have similar age structures, the bias is likely minimal. However, if some provinces have disproportionately young or elderly populations, the total population denominator may not accurately reflect the number of school-age children. A province with a larger elderly population, for instance, may seem to have less schools per capita than it actually does in relation to school-age demand, potentially biasing the coefficient downward. On the other hand, provinces with a very young population might appear to have better school coverage than they actually do. Ideally, measures of school per capita should be calculated using age-dependent population data (e.g., the actual number of primary school-age children) rather than overall population figures.

6.1.4 Schools Per Area

Although this approach intends to capture how densely schools are distributed in areas where people live, it may not accurately reflect true accessibility, particularly in geographically diverse regions. In mountainous provinces, for example, households are often scattered across uneven terrain, making travel to school more difficult. So even if the homestead land area appears larger than in flatter regions, the physical difficulty of reaching a school remains. As a result, this ratio may understate the real challenges of accessibility, since it does not take into account how scattered communities are. Total land area—though not perfect—might still offer a more consistent proxy for these spatial realities if the aim is to understand physical access or travel distance to schools. Therefore, some of the variation we observe in schools per area may reflect not true differences in school availability, but rather the influence of geography and settlement patterns. These measurement errors contribute significantly to bias in the standard errors, making the regression results less robust and causing noticeable fluctuations across different model specifications.

6.2 Models Limitation

While the fixed effects model helps mitigate certain sources of omitted variable bias, it has its own drawback. Province fixed effects control for time-invariant differences across provinces—such as long-standing economic disparities or persistent cultural norms—while year fixed effects account for nationwide shocks or trends, such as recessions or education reforms. These control methods, meanwhile, cannot fully reflect time-varying differences across provinces. For example, although a bigger city like Da Nang often provides better public services compared to a province like Son La, the difference could change over time. Provincial fixed effects, which assume such differences are constant, cannot clarify these evolving dynamics. Likewise, time fixed effects presume that all provinces respond in the same manner to national shocks, though this is not the case in reality most of the time. Even though time-varying provincial data—such as GDP per capita or unemployment rates—would help address these differences more efficiently, such data were not available for this study. The independent variables—specifically, the number of schools, schools per capita, and schools per area—also contain some limitations within themselves. These variables remain relatively stable over time within each province, which may reduce the efficiency of fixed effects models and further complicate the identification of true relationships with enrollment outcomes.

6.3 Conclusion

In conclusion, although this study has its own shortcomings, it provides a foundation for future studies to explore more effective ways to measure school accessibility and further investigate its relationship with enrollment rates in Vietnam. Future research can provide more accurate insights by tackling the measurement and model constraints as well as considering innovative approaches. These improvements will guide policymakers in creating effective, context-specific strategies to enhance educational access and outcomes across the country.

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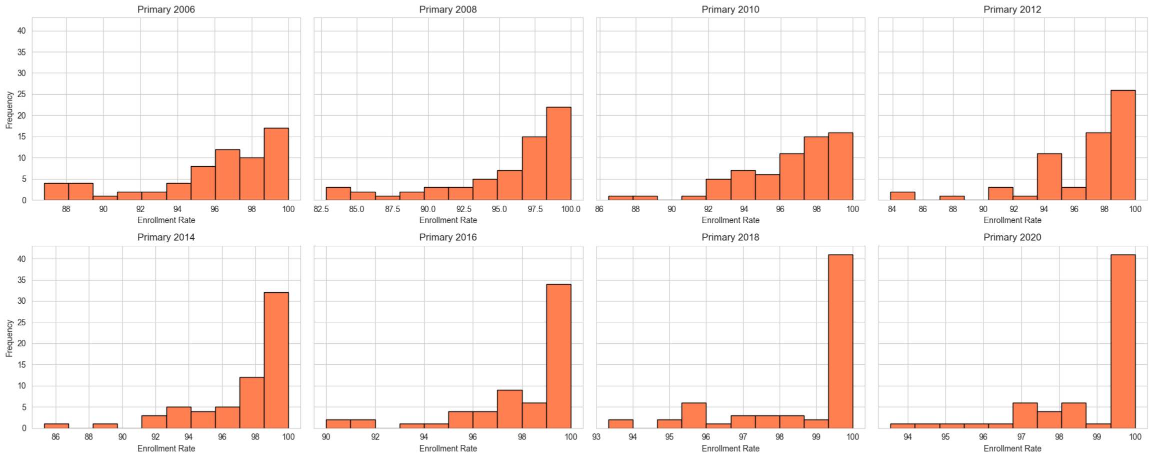
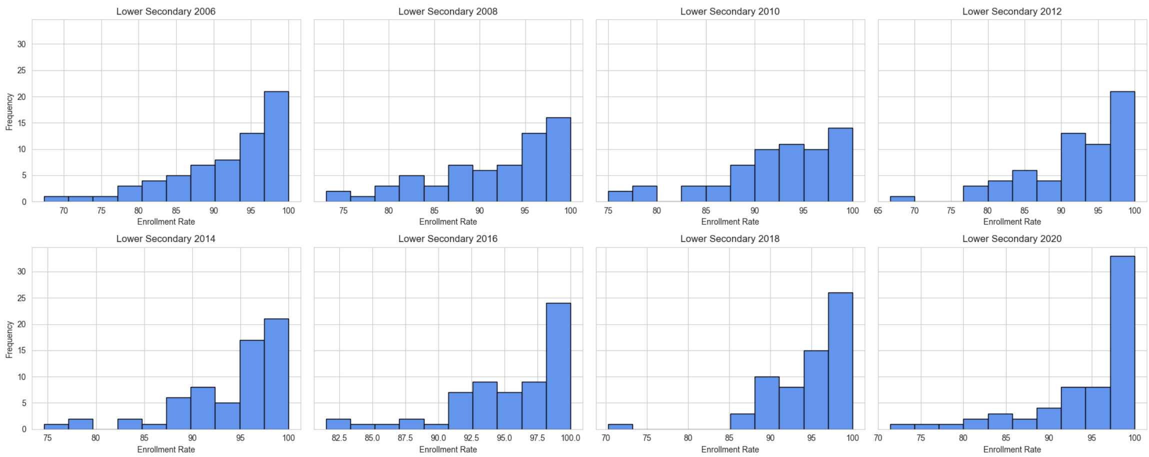
A Appendix A: Figures of Trends and Distributions

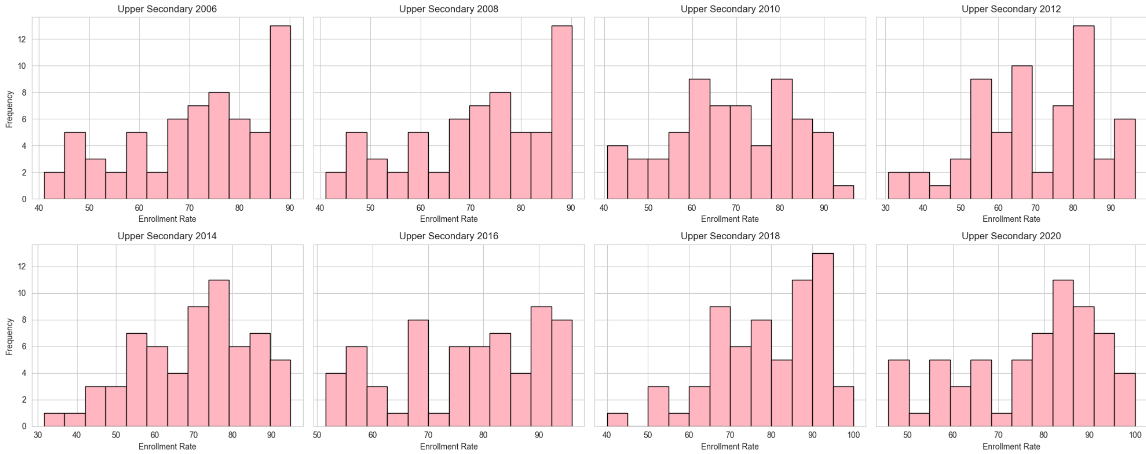
Figure A.1: *Distribution of Net Enrollment Rate across Provinces in Vietnam (2006 – 2020)*



(a) Primary Net Enrollment Rate

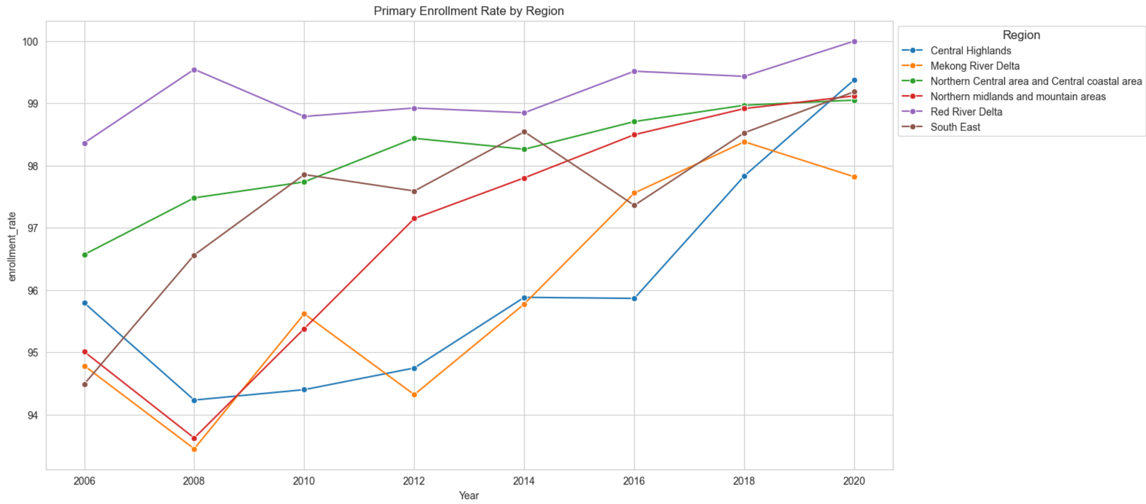


(b) Lower Secondary Net Enrollment Rate

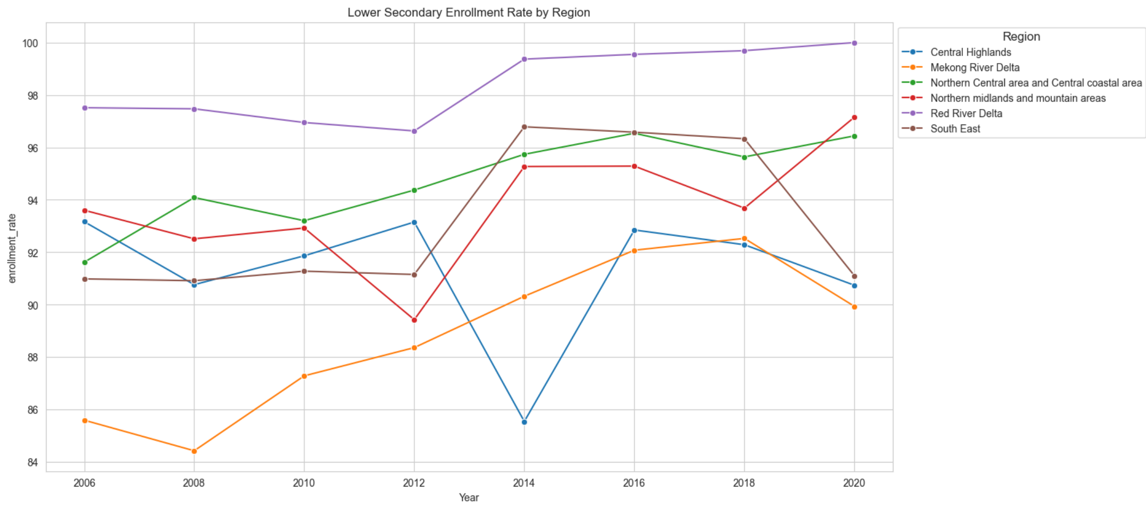


(c) Upper Secondary Net Enrollment Rate

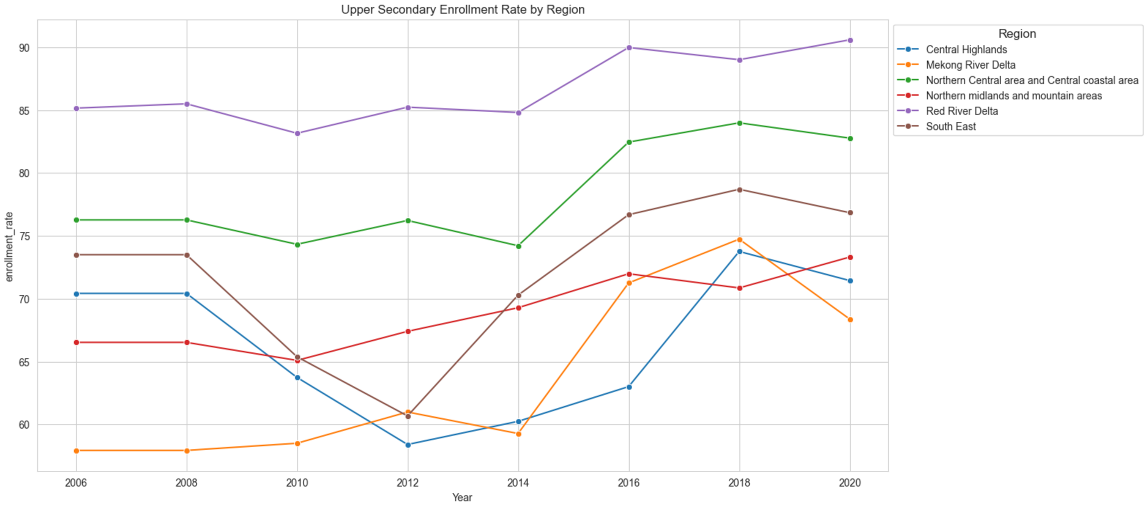
Figure A.2: *Trends in School Enrollment Rates by Region in Vietnam (2006 – 2020)*



(a) Primary Net Enrollment Rate

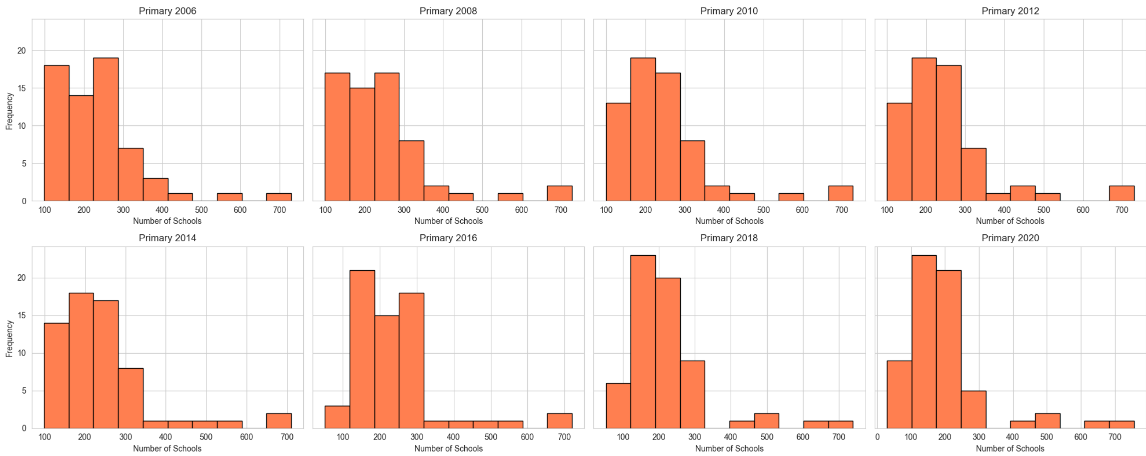


(b) Lower Secondary Net Enrollment Rate

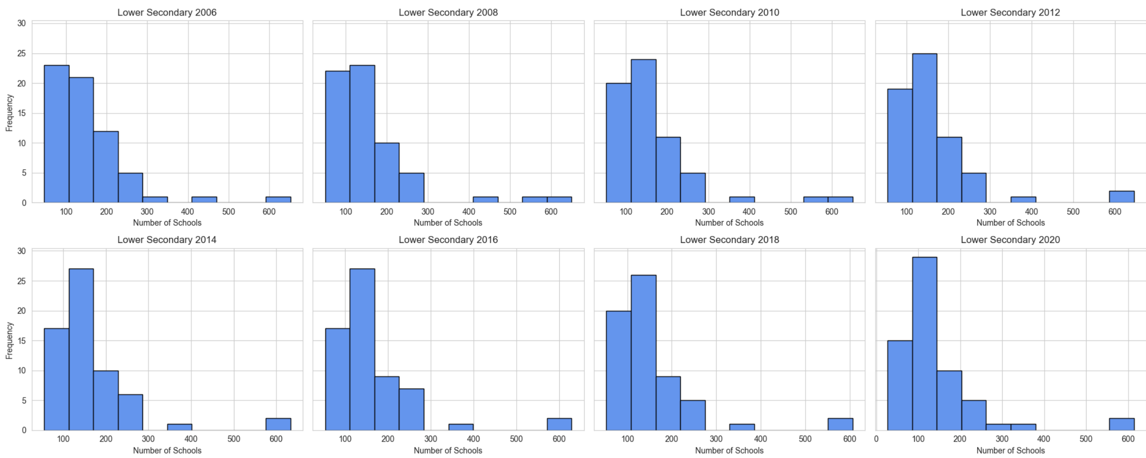


(c) Upper Secondary Net Enrollment Rate

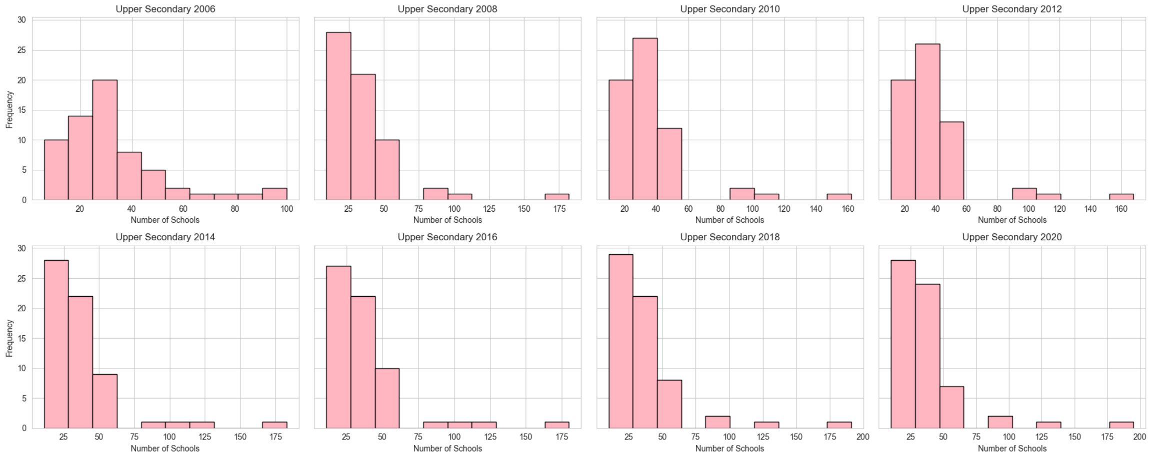
Figure A.3: *Distribution of Number of Schools across Provinces in Vietnam (2006 – 2020)*



(a) Number of Primary Schools

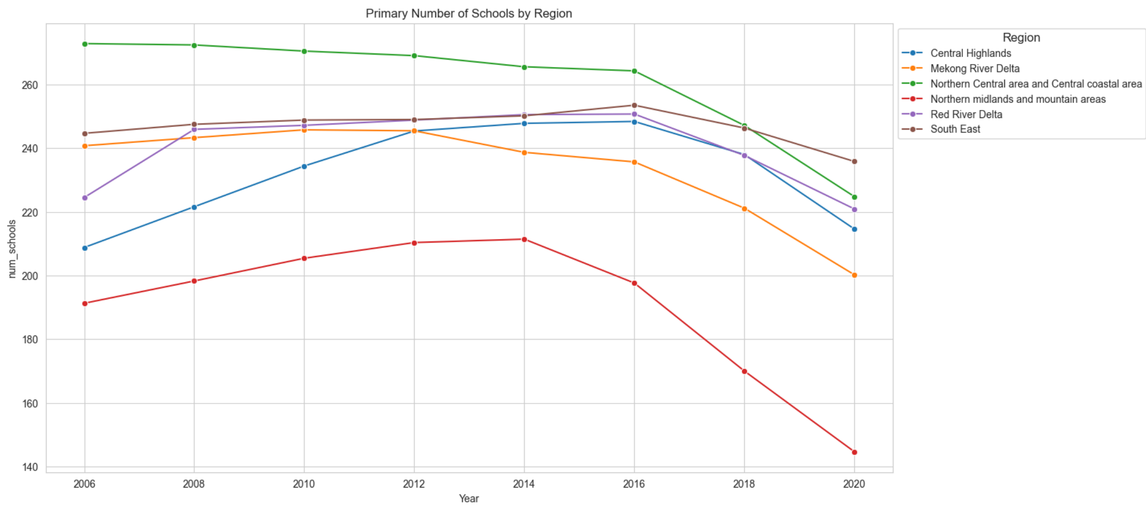


(b) Number of Lower Secondary Schools

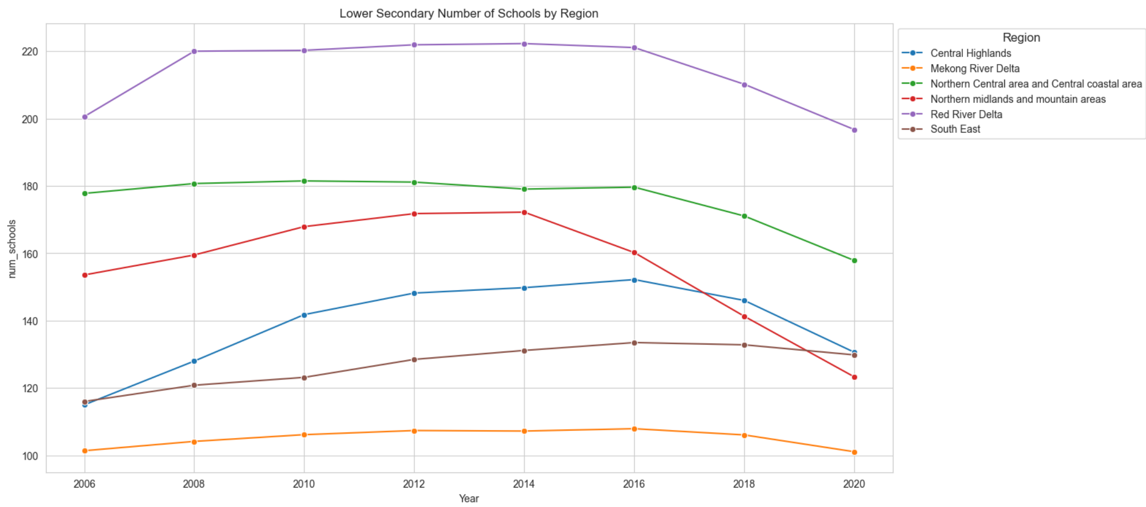


(c) Number of Upper Secondary Schools

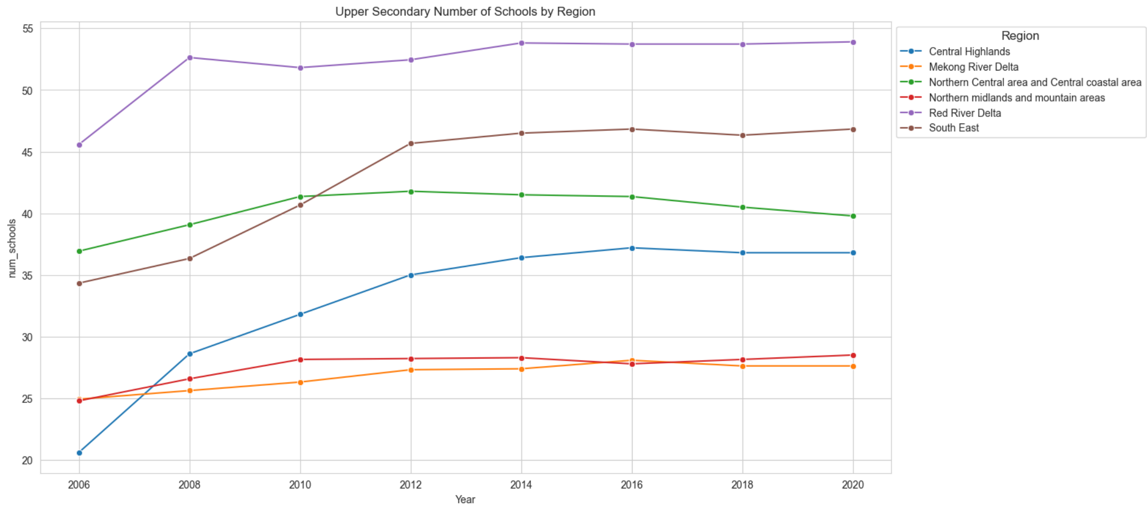
Figure A.4: *Trends in the Number of Schools across Provinces in Vietnam (2006 – 2020)*



(a) Number of Primary Schools



(b) Number of Lower Secondary Schools



(c) Number of Upper Secondary Schools